UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (10 MARKS)

Answer ALL questions.

1. Which item is NOT a programming paradigm?
   A. Structured programming
   B. Procedural programming
   C. Object-oriented programming
   D. Debugging

2. What is displayed by the following code?

```
for (int i = 1; i <= 4; i++)
    cout << i * 2;
```

   A. 2468
   B. 1357
   C. 246
   D. 1234

3. Which identifier is valid in C++?
   A. total score
   B. 2total
   C. int
   D. total_score

4. Which of the following are selection structures?
   i. if
   ii. switch
   iii. for
   iv. while
   A. i and ii
   B. ii and iv
   C. i and iii
   D. iii and iv

5. Which of the following are valid identifiers?
   i. total_score
   ii. 2total
   iii. _count
   iv. int
   A. iii and iv
   B. i and ii
   C. ii and iv
   D. i and iii

6. Which operator reads data from cin?
   A. %
   B. =
   C. >>
   D. <<

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7. Given the declaration, what is displayed?

```
char word[6] = {'p', 'r', 'i', 'n', 't', '\0'};
cout << word[3];
```

A. n
B. p
C. t
D. i

8. How many times does a while loop execute when its condition is false initially?
A. Two
B. One
C. Zero
D. Until the user stops it

9. A computer system's hardware includes which of the following?
i. input devices
ii. output devices
iii. central processing unit
iv. main memory
v. secondary storage
A. i, ii, iii, iv
B. i, ii, iii, v
C. i, iii, iv
D. i, ii, iii, iv, v

10. What is the index of the first element of a C++ array?
A. -1
B. 0
C. 1
D. The array size

PART B (60 MARKS)

QUESTION 1

This question tests the program development life cycle.

a) Define source code and object code.

(3 marks)

b) Describe THREE stages of the program development life cycle.

(2 marks)

QUESTION 2

A program reads a student's score from 0 to 100 and displays the grade shown in Table 1.

Table 1. Grade Scale

Score	Grade
80-100	A
60-79	B
50-59	C
40-49	D
0-39	F

a) Draw a flowchart for the grade classification.

(6 marks)

b) Write pseudocode for the algorithm and convert it into a complete C++ program.

(7 marks)

QUESTION 3

This question tests operator precedence and assignment.

a) Evaluate the following expressions using the declarations in the code block.

```
int a = 12, b = 5;
double c = 2.5;
```

(6 marks)

b) Write C++ statements to declare two integers a and b, read their values, and display their sum, difference, product and remainder.

(5 marks)

QUESTION 4

Study the following nested loop:

```
for (int i = 1; i <= 3; i++) {
    for (int j = 1; j <= i; j++) {
        cout << j;
    }
    cout << endl;
}
```

a) Trace the program and show the exact output.

(10 marks)

b) Identify TWO logic errors in a version that prints four rows but reuses j without resetting it. Rewrite the corrected code.

(5 marks)

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QUESTION 5

A shop stores the item prices shown in Table 1.

a) Declare an array named prices using the values in Table 1. Display all values.

Table 2. Item Prices

Item	Price (RM)
1	20.50
2	55.00
3	15.90
4	75.00
5	10.00

(6 marks)

b) Write a program segment that applies a 10% discount to prices above RM50 and stores the results in a second array. Display both arrays.

(10 marks)

PART C (30 MARKS)

QUESTION 1

A lecturer records the marks of ten students in an array.

Write a complete C++ program that reads ten marks, displays the average, the highest mark and its index, and counts how many marks are above the average.

(15 marks)

QUESTION 2

A catering company charges the meal prices shown in Table 1.

Table 3. Meal Price

Quantity Range	Price per Meal (RM)
50 - 99	11.50
100 - 199	10.50
200 and above	9.50

Write a complete C++ program that reads the customer name and quantity, validates the quantity, calculates the total, displays a receipt, and repeats until the user chooses to stop. At the end, display the total quantity and total sales.

(15 marks)

END OF QUESTION PAPER

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